

# Python for Data Analyst – Interview Questions and Answers

## ***1. Why is Python used in Data Analysis?***

Python is widely used because it has powerful libraries like Pandas, NumPy, Matplotlib, and Seaborn that make data manipulation, analysis, and visualization easy.

## ***2. What is NumPy?***

NumPy is a Python library used for numerical computing. It provides support for arrays, matrices, and mathematical functions.

## ***3. What is Pandas?***

Pandas is a data analysis library used for data manipulation and analysis using DataFrame and Series structures.

## ***4. What is a DataFrame?***

A DataFrame is a two-dimensional table-like data structure in Pandas with rows and columns.

## ***5. What is a Series in Pandas?***

A Series is a one-dimensional labeled array capable of holding different types of data.

## ***6. How do you read a CSV file in Python?***

You can read a CSV file using Pandas with the function: `pandas.read_csv()`.

## ***7. How do you handle missing values in Pandas?***

You can use functions like `dropna()`, `fillna()`, or `replace()` to handle missing values.

## ***8. What is data cleaning?***

Data cleaning is the process of removing errors, duplicates, and missing values to improve data quality.

## ***9. What is data visualization?***

Data visualization is the graphical representation of data using charts, graphs, and plots.

## ***10. Which libraries are used for data visualization in Python?***

Common libraries include Matplotlib, Seaborn, and Plotly.

**11. What is the difference between loc and iloc in Pandas?**

loc is label-based indexing while iloc is integer position-based indexing.

**12. How do you find duplicate values in a dataset?**

You can use the duplicated() function in Pandas.

**13. How do you remove duplicate rows?**

You can remove duplicates using drop\_duplicates() in Pandas.

**14. What is groupby in Pandas?**

groupby is used to split data into groups and perform aggregation operations like sum, mean, or count.

**15. What is EDA (Exploratory Data Analysis)?**

EDA is the process of analyzing datasets to summarize their main characteristics using statistics and visualization.

**16. What is a pivot table in Pandas?**

A pivot table summarizes data by grouping and aggregating values similar to Excel pivot tables.

**17. What is the use of describe() in Pandas?**

describe() generates summary statistics such as mean, count, standard deviation, min, and max.

**18. How do you filter data in Pandas?**

You can filter data using conditional statements like df[df['column'] > value].

**19. What is data wrangling?**

Data wrangling is the process of transforming raw data into a usable format for analysis.

**20. What is the difference between merge and join in Pandas?**

merge combines datasets based on common columns while join combines datasets based on index.